

Exercise 3 (18 points) – individual work

- The answers can be typed or handwritten (handwriting must be clear and readable), in this exercise sheet or your own sheet (put your name & ID at the top of the sheet). All answers must be saved to only 1 PDF file.
 - Some questions also require the submission of processes/workflows (file.rmp or file.ipynb).
 - In case of re-submission (after first grading) or submission after solution is given, your points will be weighted by 0.5.
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Record	A	B	C	Buy (class)
1	T	T	positive	yes
2	F	T	negative	no
3	T	F	neutral	yes
4	T	F	negative	no
5	F	T	neutral	no
6	T	T	positive	yes
7	F	T	negative	no
8	T	F	positive	yes
9	F	F	negative	no
10	T	T	neutral	yes

1. (Total 5 points) Use training data in the given table for Naïve Bayes classification.

Predict a target record (F, F, positive):

$$P(\text{yes}) = 0.5, P(\text{no}) = 0.5$$

$$P(A=F | \text{yes}) = 0/5 = 0$$

$$P(B=F | \text{yes}) = 2/5 = 0.4$$

$$P(C=\text{positive} | \text{yes}) = 3/5 = 0.6$$

$$P(\text{yes} | X) = (0 * 0.4 * 0.6) * 0.5 = 0$$

$$P(A=F | \text{no}) = 4/5 = 0.8$$

$$P(B=F | \text{no}) = 2/5 = 0.4$$

$$P(C=\text{positive} | \text{no}) = 0$$

$$P(\text{no} | X) = (0.8 * 0.4 * 0) * 0.5 = 0$$

With Laplace correction

$$P(\text{yes}) = 0.5, P(\text{no}) = 0.5$$

$$P(A=F | \text{yes}) = 0/5 \Rightarrow 1/7$$

$$P(B=F | \text{yes}) = 2/5 \Rightarrow 3/7$$

$$P(C=\text{positive} | \text{yes}) = 3/5 = 4/8$$

$$P(\text{yes} | X) = 0.0306 * 0.5 = 0.0153$$

$$P(A=F | \text{no}) = 4/5 \Rightarrow 5/7$$

$$P(B=F | \text{no}) = 2/5 \Rightarrow 3/7$$

$$P(C=\text{postive} | \text{no}) = 0/5 \Rightarrow 1/8$$

$$P(\text{no} | X) = 0.0383 * 0.5 = 0.019$$

1.1 (2 points) Without Laplace correction

(a) Normalized probability to predict "yes" (answer in 2 decimal places) = 0

(b) Normalized probability to predict "no" (answer in 2 decimal places) = 0

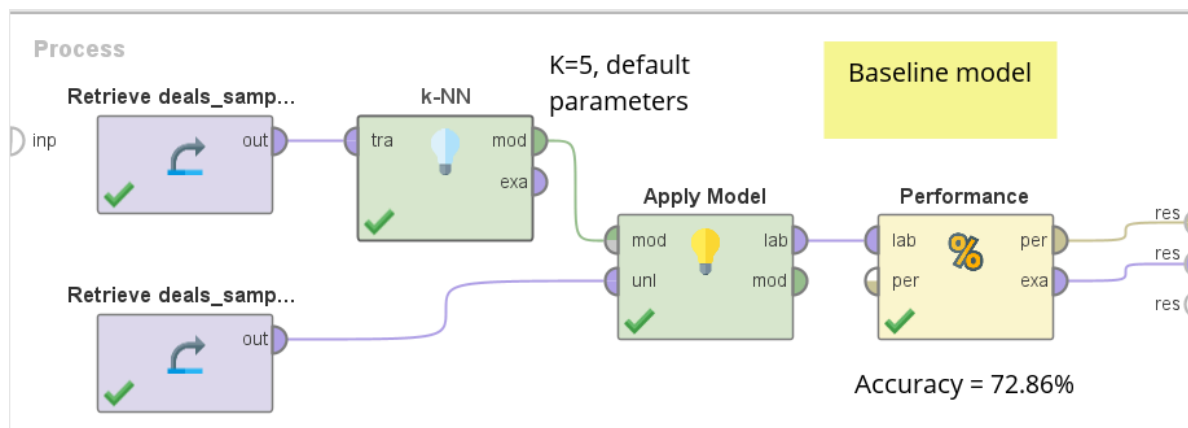
1.2 (3 points) With Laplace correction

(a) Normalized probability to predict "yes" (answer in 2 decimal places) = 0.45

(b) Normalized probability to predict "no" (answer in 2 decimal places) = 0.55

(c) Final prediction = no

For questions 2 and 3, retrieve **deals-samp-train** for training and **deals-samp-test** for testing. Consider the base line model that uses KNN with default parameters ($K=5$) to predict future customer (yes or no). Its accuracy = 72.86%.



2. (Total 4 points) Built your workflow that also uses KNN to predict future customer. But add necessary preprocessing and adjust parameters to get better accuracy than the baseline.

2.1 (1 point) Submit your workflow. It must output at least performance table. Name the workflow **question2.rmp**.

2.2 (3 points) Answer whether each attribute requires any preprocessing. If it does, briefly explain which preprocessing is applied to it.

Attribute	Preprocessing? (YES + explain, or NO)
(a) Age	yes, because it's the only numeric attribute on the data set which use different calculation of distance and similarity to nominal attribute(only 0,1) which is why we need to normalize age to make prediction not bias toward numeric value only. Apply Standardization(z transformation)
(b) Gender	no
(c) Payment Method	yes , nominal value with 3 possible value, could benefits from one hot encoding => better distance & similarity than just same or different Apply one hot encoding

3. (Total 9 points) Use neural network and SVM to classify future customer instead. Apply necessary preprocessing such that attributes are suitable for both classifiers (assuming the same preprocessing pipeline for both classifiers). Adjust parameters to get better accuracy than the baseline.

3.1 (2 points) Submit your workflow(s) for both classifiers. The workflow for each classifier must output at least performance table and classification model. Name the workflow **question3.rmp**.

3.2 (3 points) Answer whether each attribute requires any preprocessing. If it does, briefly explain which preprocessing is applied to it.

Attribute	Preprocessing? (YES + explain, or NO)
(a) Age	Yes for the same reason as question 2 because other two features are standardized and the value are on different scale than age so we need to Standardized them first Apply standardization (Z-transform)
(b) Gender	Yes, both ANN and SVM require numeric value, to compute the weight and bias.

